

Production Networks and Formation of Aggregate Uncertainty

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September 4, 2026

Abstract

JEL Codes: Keywords:

1 Benchmark Model

In the benchmark model, all agents have full information with rational expectations (FIRE) at the time when a decision is made. Time is discrete and indexed by $t = 0, 1, 2, \dots$. There is a monopolistically competitive representative firm in each sector $i \in \{1, 2, \dots, N\}$, a representative household with unit mass, competitive financial intermediaries, and a monetary authority.

Firms The monopolistically-competitive firm in each sector i (firm i , for simplicity) produces a differentiated good $y_{i,t}$ using a Cobb-Douglas production technology with labour $n_{i,t}$ and intermediate inputs $x_{ij,t}$ from other sectors j .

At the beginning of the period, the firm sets its nominal price $p_{i,t}$ before productivity $A_{i,t}$ realizes. After $A_{i,t}$ realizes, the firm determines its input demands $n_{i,t}$ and $x_{ij,t}$ and its productivity-contingent working capital $b_{i,t}$ to borrow from the financial intermediary. After the sales are realized, the firm repays the working capital $b_{i,t}$ with interest rate i_t^L . If the revenue is not sufficient to repay the working capital, the firm earns zero profit. If the firm earns positive profit, it distributes the profit to the household. Firms take wages, input prices, aggregate demand, and the loan schedule as given when choosing their own preset price.

The firm produces output according to

$$y_{i,t} = A_{i,t} n_{i,t}^{\alpha_{i,t}} \prod_{j=1}^N x_{ij,t}^{\omega_{ij,t}}, \quad (1.1)$$

where $\alpha_{i,t}$ is the labour share, and $\omega_{ij,t}$ is the cost share of intermediate input j in the production of good i . The labour and intermediate input costs are financed by the financial intermediary through working capital loans:

$$W_t n_{i,t} + \sum_{j=1}^N p_{j,t} x_{ij,t} \leq b_{i,t}. \quad (1.2)$$

The stochastic productivity process is

$$\log A_{i,t} = \bar{a}_i + \sigma_{i,t} \epsilon_{i,t}. \quad (1.3)$$

Here $\bar{a}_i$ is sector-specific average log productivity, $\sigma_{i,t}$ is the standard deviation scale of log productivity, and $\epsilon_{i,t}$ is the level shock. Let

$$\boldsymbol{\epsilon}_t = (\epsilon_{1,t}, \dots, \epsilon_{N,t})', \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_{\epsilon,t}).$$

For each i , $\epsilon_{i,t}$ has mean zero and unit variance. It realizes together with the productivity $A_{i,t}$.

Uncertainty shocks are independent across sectors and independent of level shocks. The uncertainty process is

$$\log \sigma_{i,t}^2 = (1 - \phi) \log \bar{\sigma}_i^2 + \phi \log \sigma_{i,t-1}^2 + v_{i,t}, \quad (1.4)$$

where $\phi \in (0, 1)$ and $v_{i,t} \sim \mathcal{N}(0, \sigma_v^2)$. It realizes at the very beginning of this period, before $A_{i,t}$ and $\epsilon_{i,t}$ realizes so that all economic activity can be affected by the volatility shocks contemporaneously. Due to this timing issue, define the information set until uncertainty in period t , $(\sigma_{i,t})$ realizes as $\mathcal{I}_t$ and the information set after the productivity ($A_{i,t}$, or equivalently, $\epsilon_{i,t}$) realizes as $\mathcal{F}_t$. We will always be explicit on which information set an agent is using for computing conditional moments.

This decomposition of $\sigma_{i,t}$ and $\epsilon_{i,t}$ separates level shocks from uncertainty shocks. A shock to $\epsilon_{i,t}$ changes realized productivity, while a shock to $\sigma_{i,t}$ changes the distribution of productivity conditional on $\mathcal{I}_t$.

The nominal profit of firm i is given by

$$\Pi_{i,t}^y = \max \{ p_{i,t} y_{i,t} - (1 + i_t^L) b_{i,t}, 0 \}, \quad (1.5)$$

where $p_{i,t}$ is the firm's nominal preset price. Firms choose the price first before the productivity realizes, and their labour inputs, intermediate inputs, and working capital after the productivity realizes. That is, firms choose prices to maximize the expected profit conditional on the information set $\mathcal{I}_t$ and choose the working capital and inputs to maximize the expected profit conditional on the information set $\mathcal{F}_t$.

Households Households consume a consumption aggregator C_t , make nominal deposits D_t , and supply labour N_t . They make the deposit at the end of each period t , and gain an interest at nominal gross rate $1 + i_t$ in period $t + 1$.

The utility is given by

$$U(C_t, N_t) = \log C_t - \frac{\chi}{1 + \varphi} N_t^{1+\varphi}, \quad (1.6)$$

with $\chi > 0$ and $\varphi > 0$. The consumption aggregator is a CES aggregator of differentiated goods $\{c_{i,t}\}_{i=1}^N$:

$$C_t = \left(\sum_{i=1}^N \rho_i^{\frac{1}{\xi}} c_{i,t}^{\frac{\xi-1}{\xi}} \right)^{\frac{\xi}{\xi-1}},$$

where $\rho_i \in (0, 1)$ is the demand weight of good i , $\sum_{i=1}^N \rho_i = 1$, and $\xi > 0$ is the elasticity of substitution across goods.

Households are subject to the nominal budget constraint

$$P_t C_t + D_t \leq W_t N_t + (1 + i_{t-1}) D_{t-1} + \Pi_t^y + \Pi_t^f, \quad (1.7)$$

where W_t is the nominal wage, i_{t-1} is the nominal interest rate on deposits made in period $t-1$, Π_t^y is total nominal firm profit, and Π_t^f is total nominal profit of the financial intermediary. The intermediary's profit is allowed to be negative, reflecting losses from firm defaults.

Financial Intermediary Competitive financial intermediaries take the deposit rate as tied to the policy rate, i_t . Period- t working-capital loans are funded by outstanding deposits D_{t-1} and are subject to an interest rate of $i_{i,t}^L$. Interest rates are dated by when they are set, but need not apply over the same window. The policy rate i_t is set at the beginning of period t and remunerates deposits from the beginning of period $t+1$ to the end of period $t+1$. Hence deposits D_t earn i_t in period $t+1$. The intra-period loan rate $i_{i,t}^L$ is set at the beginning of period t and applies to working-capital loans within period t . It is priced off the predetermined funding rate i_{t-1} that prevails over the same within-period window.

The loan rate is given by

$$i_{i,t}^L = i_{t-1} + s_{i,t}, \quad (1.8)$$

where the credit spread satisfies

$$s_{i,t} = \bar{s} + \psi V_t, \quad (1.9)$$

for some $\psi > 0$. The aggregate uncertainty measure is a weighted average of firm-level uncertainty measures, in the spirit of (?) and (?):

$$V_t := \sum_{i=1}^N \omega_{i,t-1}^R \sigma_{i,t-1}^2. \quad (1.10)$$

The weights are lagged nominal sales shares $\omega_{i,t-1}^R = \frac{\mathcal{R}_{i,t-1}}{\sum_{k=1}^N \mathcal{R}_{k,t-1}}$. Using lagged sales shares keeps V_t predetermined when loan rates are set.

The intermediary offers firm i a borrowing capacity $\bar{b}_{i,t}$ according to the reduced-form capacity rule

$$\bar{b}_{i,t} = \kappa \mathcal{R}_{i,t-1} \exp\left(-\frac{1}{2} \sigma_{i,t}^2 - \theta \sigma_{i,t}\right), \quad (1.11)$$

where $\kappa > 0$ captures pledgeability and $\theta > 0$ captures the sensitivity of credit capacity to firm-level uncertainty. This constraint is interpreted as a reduced-form credit-line rule based on past observable repayment capacity as an indicator of repayment ability. The term involving $\sigma_{i,t}$ captures the idea that higher conditional volatility tightens ex-ante credit capacity. Under this rule, when firms determine realized working capital after productivity is observed, their borrowing is subject to the capacity constraint

$$b_{i,t} \leq \bar{b}_{i,t}. \quad (1.12)$$

Let the lending amount of the intermediary to firm i be $\ell_{i,t}$. The intermediary's balance-sheet constraint is

$$\sum_{i=1}^N \ell_{i,t} \leq D_{t-1}. \quad (1.13)$$

The intermediary's realized repayment from firm i is $\min\{p_{i,t} y_{i,t}^d, (1 + i_{i,t}^L) b_{i,t}\}$. The realized profit of the intermediary is

$$\Pi_t^f = \sum_{i=1}^N \left[\min\{p_{i,t} y_{i,t}^d, (1 + i_{i,t}^L) b_{i,t}\} - (1 + i_{t-1}) b_{i,t} \right]. \quad (1.14)$$

Note that this profit can be negative in the case of default from many firms. All the profit or loss is borne by households.

Monetary Authority The monetary authority sets the nominal interest rate i_t at the beginning of each period. This rate applies from the beginning of period $t + 1$ to its end, equivalently, deposits D_t earn i_t in period $t + 1$. Period- t loans are priced off the predetermined funding rate i_{t-1} through (1.8). In the benchmark model, the nominal interest rate process is exogenous. An endogenous policy rule can be introduced when discussing policy implications.

1.1 Equilibrium

We now define what an equilibrium is in our benchmark model. First, note that the market-clearing conditions for goods and labour markets are

$$y_{i,t} = y_{i,t}^d := c_{i,t} + \sum_{j=1}^N x_{ji,t}, \quad (1.15)$$

$$N_t = \sum_{i=1}^N n_{i,t}, \quad (1.16)$$

The financial market clears when each firm's realized working-capital requirement is financed by the financial intermediaries subject to the credit capacity constraint:

$$\mathbb{E} [b_{i,t} \mid \mathcal{I}_t] \leq \ell_{i,t} \leq \bar{b}_{i,t} := \kappa \mathcal{R}_{i,t-1} \exp \left(-\frac{1}{2} \sigma_{i,t}^2 - \theta \sigma_{i,t} \right). \quad (1.17)$$

Before formally define the equilibrium, let us summarize the allocations, prices, and policies for clearer notion of equilibrium. The allocations are stochastic sequences of

$$\mathfrak{A} = \left\{ \left\{ c_{i,t}, n_{i,t}, y_{i,t}, b_{i,t}, \ell_{i,t}, \{x_{ij,t}\}_{j=1}^N \right\}_{i=1}^N, C_t, N_t, D_t \right\}_{t=0}^{\infty}.$$

The prices are

$$\mathfrak{P} = \left\{ \{p_{i,t}\}_{i=1}^N, P_t, W_t \right\}_{t=0}^{\infty}.$$

The policy variables are

$$\mathfrak{M} = \{i_{t-1}, i_t^L, s_t\}_{t=0}^{\infty}.$$

Note that we do not explicitly define the policy variables for the monetary authority because we are not yet there to discuss the monetary policy.

Definition 1 (Benchmark Model Equilibrium). *A general equilibrium is a triplet $(\mathfrak{A}, \mathfrak{P}, \mathfrak{M})$ such that the following conditions hold:*

- (a) *the monopolistically-competitive firms in each industry set prices $p_{i,t}$ to maximize the expected profit given the uncertainty information and optimally choose the working capital and inputs to meet the realized productivity-contingent demand;*
- (b) *the representative household chooses consumption and labour supply to maximize utility subject to its budget constraint;*
- (c) *the financial intermediary sets the loan rate i_t^L and borrowing capacity $\ell_{i,t}$ to maximize its expected profit subject to the balance-sheet constraint and the borrowing-capacity constraint;*
- (d) *all markets clear.*

In the following, we will only present equilibrium results that matter for uncertainty and network discussions. Other equilibrium conditions will be presented in Appendix I.1.

We now define the notion of production network. Recall that for firm i , the cost share of input j is $\omega_{ij,t}$ as defined in the production function (1.1). The production network is an $N \times N$ matrix with the (i, j) -th element defined by the normalized cost share:

$$\tilde{\omega}_{ij,t} := \frac{\omega_{ij,t}}{1 - \alpha_{i,t}},$$

so that the row sum of $\tilde{\omega}_{ij,t}$ equals 1. We will denote the matrix with $\omega_{ij,t}$ entries by Ω_t and the matrix with $\tilde{\omega}_{ij,t}$ entries by $\tilde{\Omega}_t$.

1.1.1 Optimal Prices

Recall that firms choose prices after uncertainty $\sigma_{i,t}$ realizes but before the level shock $\epsilon_{i,t}$ realizes. The optimal pricing problem is to maximize expected profit conditional on the information set $\mathcal{I}_t$. For tractability, we formulate the price-setting problem as an unconstrained problem. The borrowing-capacity constraint is not imposed directly at the price-setting stage because the realized working-capital requirement $b_{i,t}$ depends on the subsequent productivity

realization. Instead, firms choose prices taking the loan rate and limited-liability payoff as given, through which the borrowing friction enters expected profits.

We can derive that the price of good i should satisfy the following equation:

$$p_{i,t} = \left(1 + \frac{1}{(\xi - 1)v_{i,t}}\right) (1 + i_t^L) \mu_{i,t|I_t}^{A^{-1}} \frac{\Phi(q_{i,t} - \sigma_{i,t})}{\Phi(q_{i,t})} mc_{i,t}, \quad (1.18)$$

where $v_{i,t}$ is the final consumption fraction of total sales, $\mu_{i,t}^{A^{-1}} = e^{-\bar{a}_i + \frac{1}{2}\sigma_{i,t}^2}$ is the conditional mean of the inverse of productivity $\mathbb{E}[A_{i,t}^{-1}|I_t]$, and $q_{i,t}$ is the productivity cutoff of default such that firm i does not default if and only if the standardized productivity $A_{i,t}$ is greater than or equal to $-q_{i,t}$. Details are in Appendix I.1.2.

Every component in equation (1.18) has its explicit interpretation. First, the monopoly markup is $1 + \frac{1}{(\xi-1)v_{i,t}}$. Note that when $v_{i,t} \rightarrow 1$, that is, no intermediate input demand, we have this term converges to $\frac{\xi}{\xi-1}$, which is the standard CES markup. Second, the financial wedge is $1 + i_t^L$, reflecting the interest cost of borrowing. Third, the uncertainty wedge is brought by $\mu_{i,t|I_t}^{A^{-1}}$ which increases with uncertainty $\sigma_{i,t}$, meaning that the distortion is larger when the uncertainty is larger. Finally, the normal CDF ratio represents the default/liquidity-risk adjustment. Consider two cases. When $q_{i,t} \rightarrow +\infty$, it is very unlikely that the firm will default. This adjustment term goes to 1 so that this wedge vanishes. On the other hand, when $q_{i,t} \rightarrow -\infty$, default becomes almost certain. The firm repays only in exceptionally good productivity states, where producing the required output is very cheap. Therefore, the marginal cost conditional on repayment becomes negligible, which pushes the optimal price toward zero.

We will show the existence and uniqueness of the equilibrium prices after characterizing the complete general equilibrium.

1.1.2 Determining Working Capital

The second-stage problem of firms is cost minimization, or equivalently, profit maximizing conditional on the information set $\mathcal{F}_t$.

$$\max_{b_{i,t}, n_{i,t}, \{x_{ij,t}\}_{j=1}^N} \mathbb{E} \left[\Pi_{i,t}^y \mid \mathcal{F}_t \right], \quad (1.19)$$

subject to the working capital constraint (1.2) and the borrowing-capacity rule (1.12).

1.1.3 General Equilibrium

Now we would like to show that the equilibrium prices exist and are unique. Before that, we need a standard assumption on the production network such that the Leontief inverse matrix is well-defined.

Assumption 1.1 (Production Network Assumption). *The production network is such that $\mathcal{L}_t = I_N - \Omega_t$ is non-singular for all t .*

Now we can state the existence and uniqueness of finite equilibrium price.

Proposition 1.1 (Existence and Uniqueness of Finite Equilibrium Price). *Under Assumption 1.1, for each good i , there exists a unique finite equilibrium price $p_{i,t}$ that satisfies the optimal pricing equation (1.18).*

Proof. See Appendix I.1.4. □

I Model Derivation Details

I.1 Benchmark Model

I.1.1 Micro-foundation for Reduced-Form Financial Intermediary Constraints

The financial intermediary's optimization problem is to maximize its nominal profit subject to a default-tolerance rule:

$$\max_{\ell_{i,t}} \mathbb{E} \left[\Pi_t^f \mid \mathcal{I}_t \right],$$

subject to a default-tolerance constraint

$$\mathbb{P} \left(\mathcal{R}_{i,t} < (1 + i_t^L) \ell_{i,t} \mid \mathcal{I}_t \right) \leq \delta, \quad (\text{I.1})$$

for some $\delta \in (0, 1)$, and the balance sheet constraint (1.13).

Denote $\mu_{i,t}^A := \mathbb{E} [A_{i,t} \mid \mathcal{I}_t] = \exp \left(\bar{a}_i + \frac{1}{2} \sigma_{i,t}^2 \right)$ consider the objective function:

$$\mathbb{E} \left[\Pi_t^f \mid \mathcal{I}_t \right] = \sum_{i=1}^N \left[p_{i,t} \mu_{i,t}^A \tilde{y}_{i,t} (1 - \Phi(q_{2i,t})) + (1 + i_{i,t}^L) \ell_{i,t} \Phi(q_{1i,t}) - \ell_{i,t} \right] - i_{t-1} D_{t-1}, \quad (\text{I.2})$$

where

$$q_{1i,t} = \frac{\bar{a}_i - \log((1 + i_t^L) \ell_{i,t}) + \log(p_{i,t} \tilde{y}_{i,t})}{\sigma_{i,t}}, \quad q_{2i,t} = q_{1i,t} + \sigma_{i,t}.$$

Here we pause a bit for a technical lemma that is directly obtained from the definitions of $q_{1i,t}$ and $q_{2i,t}$.

Lemma I.1 (Densities at $q_{1i,t}$ and $q_{2i,t}$). *The probability densities of $q_{1i,t}$ and $q_{2i,t}$ have the following relationship:*

$$\frac{\phi(q_{2i,t})}{\phi(q_{1i,t})} = \frac{(1 + i_t^L) \ell_{i,t}}{p_{i,t} \mu_{i,t}^A \tilde{y}_{i,t}}.$$

Proof. Note that $q_{2i,t} = q_{1i,t} + \sigma_{i,t}$. Therefore,

$$\frac{\phi(q_{2i,t})}{\phi(q_{1i,t})} = \exp \left(-q_{1i,t} \sigma_{i,t} - \frac{1}{2} \sigma_{i,t}^2 \right).$$

Substituting the expression of $q_{1i,t}$ into the expression yields

$$\begin{aligned}\frac{\phi(q_{2i,t})}{\phi(q_{1i,t})} &= \exp\left(-\bar{a}_i + \log\left((1+i_t^L)\ell_{i,t}\right) - \log(p_{i,t}\tilde{y}_{i,t}) - \frac{1}{2}\sigma_{i,t}^2\right) \\ &= \frac{(1+i_t^L)\ell_{i,t}}{p_{i,t}\tilde{y}_{i,t}} \exp\left(-\bar{a}_i - \frac{1}{2}\sigma_{i,t}^2\right) \\ &= \frac{(1+i_t^L)\ell_{i,t}}{p_{i,t}\mu_{i,t|I_t}^A \tilde{y}_{i,t}}.\end{aligned}$$

□

Corollary I.1 (Default-tolerance constraint binds). *When the default-tolerance constraint binds, $q_{1i,t} = z_{1-\delta}$ for all i .*

Proof. Suppose that the default-tolerance constraint binds. Then we have

$$(1+i_t^L)\ell_{i,t} = p_{i,t}\mu_{i,t|I_t}^A \tilde{y}_{i,t} \exp\left(-\frac{1}{2}\sigma_{i,t-1}^2 - z_{1-\delta}\sigma_{i,t-1}\right). \quad (\text{I.3})$$

By Lemma I.1, we have

$$\exp\left(-\frac{1}{2}\sigma_{i,t-1}^2 - z_{1-\delta}\sigma_{i,t-1}\right) = \frac{\phi(q_{2i,t})}{\phi(q_{1i,t})} = \exp\left(-\frac{1}{2}\sigma_{i,t-1}^2 - q_{1i,t}\sigma_{i,t-1}\right).$$

It follows from the monotonicity of the exponential functions that $z_{1-\delta} = q_{1i,t}$. □

The derivation below shows that competitive risk-based lending implies a loan rate that is increasing in revenue risk. We use the linear spread rule (1.9) as a reduced-form approximation to this structural pricing relation.

Since financial intermediaries are perfectly competitive, they should have zero expected profit for each loan i made. The potential realized loss will be borne by household consumption. Suppose that the default-tolerance constraint binds. Then equation (I.3) and the zero-expected profit condition for loan i together imply that

$$p_{i,t}\mu_{i,t|I_t}^A \tilde{y}_{i,t} (1 - \Phi(z_{1-\delta} + \sigma_{i,t})) + (1+i_t^L)\ell_{i,t}(1-\delta) = (1+i_{t-1})\ell_{i,t}.$$

Further simplifying the expression yields

$$1+i_t^L = (1+i_{t-1}) \left[\exp\left(\frac{1}{2}\sigma_{i,t-1}^2 + z_{1-\delta}\sigma_{i,t-1}\right) (1 - \Phi(z_{1-\delta} + \sigma_{i,t-1})) + (1-\delta) \right]^{-1}$$

We now show that the multiplier to $1 + i_{t-1}$ is greater than or equal to 1. Note that

$$\begin{aligned}
& \exp\left(\frac{1}{2}\sigma_{i,t-1}^2 + z_{1-\delta}\sigma_{i,t-1}\right) (1 - \Phi(z_{1-\delta} + \sigma_{i,t-1})) \\
&= \exp\left(\frac{1}{2}\sigma_{i,t-1}^2 + z_{1-\delta}\sigma_{i,t-1}\right) \int_{z_{1-\delta} + \sigma_{i,t-1}}^{+\infty} \phi(u) du \\
&= \exp\left(\frac{1}{2}\sigma_{i,t-1}^2 + z_{1-\delta}\sigma_{i,t-1}\right) \int_{z_{1-\delta}}^{+\infty} \phi(x + \sigma_{i,t-1}) dx \\
&= \int_{z_{1-\delta}}^{+\infty} \phi(x) \exp(-\sigma(x - z_{1-\delta})) dx \\
&\leq \int_{z_{1-\delta}}^{+\infty} \phi(x) dx \\
&= \delta,
\end{aligned}$$

where the inequality is derived from $x \geq z_{1-\delta}$. The result follows.

Taking log approximation on both sides, we get

$$i_t^L = i_{t-1} + \left[\delta - \exp\left(\frac{1}{2}\sigma_{i,t-1}^2 + z_{1-\delta}\sigma_{i,t-1}\right) (1 - \Phi(z_{1-\delta} + \sigma_{i,t-1})) \right].$$

It remains to show that the second term on the right-hand side is increasing in $\sigma_{i,t-1}$. Differentiating the second term with respect to $\sigma_{i,t-1}$, we get

$$\begin{aligned}
& \frac{d}{d\sigma_{i,t-1}} \left[\delta - \exp\left(\frac{1}{2}\sigma_{i,t-1}^2 + z_{1-\delta}\sigma_{i,t-1}\right) (1 - \Phi(z_{1-\delta} + \sigma_{i,t-1})) \right] \\
&= \exp\left(\frac{1}{2}\sigma_{i,t-1}^2 + z_{1-\delta}\sigma_{i,t-1}\right) \left[\phi(z_{1-\delta} + \sigma_{i,t-1}) - (z_{1-\delta} + \sigma_{i,t-1}) (\Phi(-z_{1-\delta} - \sigma_{i,t-1})) \right].
\end{aligned}$$

Consider the function $f(a) = \phi(a) - a\Phi(-a)$. Note that

$$\begin{aligned}
f'(a) &= \phi'(a) - \Phi(-a) + a\phi(-a) \\
&= -a\phi(a) - \Phi(-a) + a\phi(a) \\
&= -\Phi(-a) < 0.
\end{aligned}$$

Moreover, by the Mills ratio,

$$\lim_{a \rightarrow +\infty} [\phi(a) - a\Phi(-a)] = 0.$$

Since $f'(a) = -\Phi(-a) < 0$, f is strictly decreasing and converges to zero from above. Therefore $f(a) > 0$ for every finite a . This implies that the second term on the right-hand

side is increasing in $\sigma_{i,t-1}$. Then we are able to define the second term as $s_{i,t} = \bar{s} + \psi V_t$ in a reduced-form representation of competitive risk-based lending.

Another way is to completely solve for the optimization problem subject to the balance sheet constraint (1.13) with Lagrange Multiplier λ_t^f . This provides the optimal lending amount of the financial intermediary, as is stated in the following Lemma.

Lemma I.2 (Optimal Lending Amount). *Suppose the intermediary chooses $\{\ell_{i,t}\}_{i=1}^N$ to maximize expected profit subject to the balance-sheet constraint and the default-tolerance constraint. Then the optimal lending amount $\ell_{i,t}$ is characterized by*

$$\Phi(q_{1i,t}) = \begin{cases} \frac{1}{1+i_t^L}, & \text{if } \sum_{i=1}^N \ell_{i,t} < D_{t-1} \text{ and } \mathbb{P}(\mathcal{R}_{i,t} < (1+i_t^L)\ell_{i,t} \mid \mathcal{I}_t) < \delta, \\ \frac{1+i_{t-1}}{1+i_t^L}, & \text{if } \sum_{i=1}^N \ell_{i,t} = D_{t-1} \text{ and } \mathbb{P}(\mathcal{R}_{i,t} < (1+i_t^L)\ell_{i,t} \mid \mathcal{I}_t) < \delta, \\ 1 - \delta, & \text{if } \mathbb{P}(\mathcal{R}_{i,t} < (1+i_t^L)\ell_{i,t} \mid \mathcal{I}_t) = \delta. \end{cases}$$

Proof. Let

$$R_i^f(\ell_{i,t}) := \mathbb{E} \left[\min \left\{ \mathcal{R}_{i,t}, (1+i_t^L)\ell_{i,t} \mid \mathcal{I}_t \right\} \right].$$

Using the lognormal formula derived above, we can write

$$R_i^f(\ell_{i,t}) = p_{i,t} \mu_{i,t|\mathcal{I}_t}^A \tilde{y}_{i,t} (1 - \Phi(q_{2i,t})) + (1+i_t^L)\ell_{i,t} \Phi(q_{1i,t}).$$

Differentiating with respect to $\ell_{i,t}$ gives

$$R_i^{f'}(\ell_{i,t}) = (1+i_t^L)\Phi(q_{1i,t}),$$

where the density terms cancel by Lemma I.1.

Case 1: Balance-sheet constraint does not bind and the default-tolerance constraint does not bind. When the default-tolerance constraint does not bind, the lending FOC is

$$R_i^{f'}(\ell_{i,t}) = 1 + \lambda_t^f,$$

where λ_t^f is the multiplier on the balance-sheet constraint

$$D_{t-1} - \sum_{i=1}^N \ell_{i,t} \geq 0.$$

Therefore,

$$(1 + i_t^L)\Phi(q_{1i,t}) = 1 + \lambda_t^f.$$

If the balance-sheet constraint does not bind, complementary slackness implies

$$\lambda_t^f = 0.$$

Hence

$$\Phi(q_{1i,t}) = \frac{1}{1 + i_t^L}.$$

Case 2: Balance-sheet constraint binds and the default-tolerance constraint does not bind. If the balance-sheet constraint binds, then

$$D_{t-1} = \sum_{i=1}^N \ell_{i,t}.$$

Therefore, the intermediary's expected profit can be rewritten as

$$\mathbb{E} \left[\Pi_t^f \mid \mathcal{I}_t \right] = \sum_{i=1}^N \left[p_{i,t} \mu_{i,t|\mathcal{I}_t}^A \tilde{y}_{i,t} (1 - \Phi(q_{2i,t})) + (1 + i_t^L) \ell_{i,t} \Phi(q_{1i,t}) - (1 + i_{t-1}) \ell_{i,t} \right].$$

Since the default-tolerance constraint does not bind in this case, the FOC is

$$R_i^{f'}(\ell_{i,t}) = 1 + i_{t-1}.$$

Using $R_i^{f'}(\ell_{i,t}) = (1 + i_t^L)\Phi(q_{1i,t})$, we obtain

$$(1 + i_t^L)\Phi(q_{1i,t}) = 1 + i_{t-1}.$$

Hence

$$\Phi(q_{1i,t}) = \frac{1 + i_{t-1}}{1 + i_t^L}.$$

Comparing this condition with the original KKT condition

$$(1 + i_t^L)\Phi(q_{1i,t}) = 1 + \lambda_t^f$$

also implies that, in this case,

$$\lambda_t^f = i_{t-1}.$$

Case 3: Default-tolerance constraint binds. The default-tolerance constraint is

$$\mathbb{P}\left(\mathcal{R}_{i,t} < (1 + i_t^L)\ell_{i,t} \mid \mathcal{I}_t\right) \leq \delta.$$

Since

$$\mathbb{P}\left(\mathcal{R}_{i,t} < (1 + i_t^L)\ell_{i,t} \mid \mathcal{I}_t\right) = 1 - \Phi(q_{1i,t}),$$

the default-tolerance constraint can be written as

$$\Phi(q_{1i,t}) \geq 1 - \delta.$$

If the default-tolerance constraint binds, then

$$\Phi(q_{1i,t}) = 1 - \delta.$$

This proves the result. $\square$

The lemma shows that optimal lending is governed by two objects. The first is the lending amount implied by the intermediary's expected-profit FOC. The second is the maximum lending amount allowed by the default-tolerance constraint. The realized optimal lending amount is the smaller of these two values.

To make this explicit, use the definition

$$q_{1i,t} = \frac{\bar{a}_i + \log(p_{i,t}\tilde{y}_{i,t}) - \log((1 + i_t^L)\ell_{i,t})}{\sigma_{i,t}}.$$

If the default-tolerance constraint does not bind, the marginal funding cost is

$$mfc_t = \begin{cases} 1, & \text{if } \sum_{i=1}^N \ell_{i,t} < D_{t-1}, \\ 1 + i_{t-1}, & \text{if } \sum_{i=1}^N \ell_{i,t} = D_{t-1}. \end{cases}$$

Then $q_{1i,t} = \Phi^{-1}\left(\frac{mfc_t}{1 + i_t^L}\right)$. The lending amount implied by the expected-profit FOC is then

$$\ell_{i,t}^u = \frac{p_{i,t}\tilde{y}_{i,t} \exp(\bar{a}_i)}{1 + i_t^L} \exp\left[-\sigma_{i,t} \Phi^{-1}\left(\frac{mfc_t}{1 + i_t^L}\right)\right].$$

Equivalently, since $\mu_{i,t|I_t}^A = \exp\left(\bar{a}_i + \frac{1}{2}\sigma_{i,t}^2\right)$, we can write

$$\ell_{i,t}^u = \frac{p_{i,t}\tilde{y}_{i,t}\mu_{i,t|I_t}^A}{1+i_t^L} \exp\left[-\frac{1}{2}\sigma_{i,t}^2 - \sigma_{i,t}\Phi^{-1}\left(\frac{mfc_t}{1+i_t^L}\right)\right].$$

If the default-tolerance constraint binds, then

$$q_{1i,t} = z_{1-\delta}.$$

The lending cap induced by the default-tolerance constraint is therefore

$$\ell_{i,t}^{\max} = \frac{p_{i,t}\tilde{y}_{i,t}\exp(\bar{a}_i)}{1+i_t^L} \exp(-z_{1-\delta}\sigma_{i,t}).$$

Equivalently,

$$\ell_{i,t}^{\max} = \frac{p_{i,t}\tilde{y}_{i,t}\mu_{i,t|I_t}^A}{1+i_t^L} \exp\left(-\frac{1}{2}\sigma_{i,t}^2 - z_{1-\delta}\sigma_{i,t}\right).$$

Hence the optimal lending amount can be written as

$$\ell_{i,t} = \min\{\ell_{i,t}^u, \ell_{i,t}^{\max}\}.$$

This expression clarifies the role of aggregate uncertainty. Since the loan rate satisfies

$$i_t^L = i_{t-1} + s_t, \quad s_t = \bar{s} + \psi V_t,$$

aggregate uncertainty affects lending through i_t^L . When the default-tolerance constraint is slack, higher V_t raises i_t^L , but the effect on $\ell_{i,t}^u$ is generally ambiguous. This is because a higher loan rate both lowers the repayment probability required by the FOC and mechanically divides the promised repayment by $1+i_t^L$. Therefore, it is not generally valid to conclude that $\ell_{i,t}^u$ increases with V_t .

When the default-tolerance constraint binds, lending is pinned down by

$$\Phi(q_{1i,t}) = 1 - \delta.$$

In this case, the constraint fixes the maximum admissible default probability, or equivalently the cutoff $q_{1i,t} = z_{1-\delta}$. The implied lending cap satisfies

$$\ell_{i,t}^{\max} = \frac{p_{i,t}\tilde{y}_{i,t}\mu_{i,t|I_t}^A}{1+i_t^L} \exp\left(-\frac{1}{2}\sigma_{i,t}^2 - z_{1-\delta}\sigma_{i,t}\right).$$

Thus, holding firm revenue fundamentals fixed, higher aggregate uncertainty V_t reduces $\ell_{i,t}^{\max}$ through the higher loan rate i_t^L . Higher firm-level uncertainty $\sigma_{i,t}$ also reduces the lending cap through the default-tolerance term.

I.1.2 Firms' Optimal Pricing

The Cobb-Douglas production function (1.1) implies the cost-share conditions, which will be formally shown in Appendix I.1.3:

$$n_{i,t} = \frac{\alpha_{i,t} b_{i,t}}{W_t}, \quad x_{ij,t} = \frac{\omega_{ij,t} b_{i,t}}{p_{j,t}}, \forall j = 1, \dots, N.$$

Therefore, we can rewrite the production function as

$$y_{i,t} = A_{i,t} b_{i,t} \underbrace{\frac{\alpha_{i,t}^{1-\alpha_{i,t}} \prod_{j=1}^N \omega_{ij,t}^{\alpha_{ij,t}}}{W_t^{\alpha_{i,t}} \prod_{j=1}^N p_{j,t}^{\alpha_{ij,t}}}}_{\equiv \frac{1}{mc_{i,t}}}.$$

At equilibrium, we have $y_{i,t} = y_{i,t}^d$ so that

$$y_{i,t}^d = A_{i,t} \frac{b_{i,t}}{mc_{i,t}} \iff b_{i,t} = mc_{i,t} \frac{y_{i,t}^d}{A_{i,t}}.$$

Substituting this into the firm's optimal pricing problem where the profit is defined as in (1.5), we get the equivalent optimization problem:

$$\begin{aligned} & \max_{p_{i,t}} \mathbb{E} \left[\Pi_{i,t}^y \mid \mathcal{I}_t \right] \\ &= \max_{p_{i,t}} \mathbb{E} \left[\max \left\{ p_{i,t} y_{i,t}^d(p_{i,t}) - (1 + i_t^L) b_{i,t}, 0 \right\} \mid \mathcal{I}_t \right] \\ &= \max_{p_{i,t}} y_{i,t}^d(p_{i,t}) \mathbb{E} \left[\max \left\{ p_{i,t} - (1 + i_t^L) mc_{i,t} \frac{1}{A_{i,t}}, 0 \right\} \mid \mathcal{I}_t \right] \\ &= \max_{p_{i,t}} y_{i,t}^d(p_{i,t}) \mathbb{E} \left[\left(p_{i,t} - (1 + i_t^L) mc_{i,t} \frac{1}{A_{i,t}} \right) \mathbb{1} \left\{ p_{i,t} > (1 + i_t^L) mc_{i,t} \frac{1}{A_{i,t}} \right\} \mid \mathcal{I}_t \right] \\ &= \max_{p_{i,t}} y_{i,t}^d(p_{i,t}) \left[p_{i,t} \Phi(q_{i,t}) - (1 + i_t^L) \mu_{i,t| \mathcal{I}_t}^{A^{-1}} mc_{i,t} \Phi(q_{i,t} - \sigma_{i,t}) \right], \end{aligned}$$

where $\mu_{i,t| \mathcal{I}_t}^{A^{-1}} = \exp \left(-\bar{a}_i + \frac{1}{2} \sigma_{i,t}^2 \right)$ is the conditional mean of productivity $A_{i,t}^{-1}$ derived by the property of log-normal distribution, and $q_{i,t}$ is the cutoff for whether the firm has zero profit:

$$q_{i,t} = \frac{\bar{a}_i + \log(p_{i,t}) - \log \left((1 + i_t^L) mc_{i,t} \right)}{\sigma_{i,t}},$$

The FOC of the optimal pricing problem is

$$\begin{aligned}
[p_{i,t}] : \frac{\partial y_{i,t}^d(p_{i,t})}{\partial p_{i,t}} & \left[p_{i,t} \Phi(q_{i,t}) - (1 + i_t^L) \mu_{i,t|I_t}^{A^{-1}} m c_{i,t} \Phi(q_{i,t} - \sigma_{i,t}) \right] \\
& = -y_{i,t}^d(p_{i,t}) \left[\Phi(q_{i,t}) + p_{i,t} \phi(q_{i,t}) \frac{\partial q_{i,t}}{\partial p_{i,t}} - (1 + i_t^L) \mu_{i,t|I_t}^{A^{-1}} m c_{i,t} \phi(q_{i,t} - \sigma_{i,t}) \frac{\partial q_{i,t}}{\partial p_{i,t}} \right].
\end{aligned} \tag{I.4}$$

By the definition of $q_{i,t}$, note that

$$\begin{aligned}
p_{i,t} & = \exp(-\bar{a}_i + q_{i,t} \sigma_{i,t}) (1 + i_t^L) m c_{i,t} \\
& = \exp\left(-\frac{1}{2} \sigma_{i,t}^2 + q_{i,t} \sigma_{i,t}\right) (1 + i_t^L) \mu_{i,t|I_t}^{A^{-1}} m c_{i,t} \\
& = (1 + i_t^L) \mu_{i,t|I_t}^{A^{-1}} m c_{i,t} \frac{\phi(q_{i,t} - \sigma_{i,t})}{\phi(q_{i,t})}.
\end{aligned}$$

Since $\frac{\partial q_{i,t}}{\partial p_{i,t}} = \frac{1}{\sigma_{i,t} p_{i,t}}$ is finite for finite $p_{i,t}$, the FOC (I.4) can be rewritten as

$$\frac{\partial y_{i,t}^d(p_{i,t})}{\partial p_{i,t}} \left[p_{i,t} \Phi(q_{i,t}) - (1 + i_t^L) \mu_{i,t|I_t}^{A^{-1}} m c_{i,t} \Phi(q_{i,t} - \sigma_{i,t}) \right] + y_{i,t}^d(p_{i,t}) \Phi(q_{i,t}) = 0.$$

Consider the demand for good i derived from the household consumption aggregator:

$$\begin{aligned}
y_{i,t}^d(p_{i,t}) & = c_{i,t} + \sum_{j=1}^N x_{ji,t} \\
& = \rho_i \left(\frac{p_{i,t}}{P_t} \right)^{-\xi} C_t + \sum_{j=1}^N \frac{\omega_{ji,t} b_{j,t}}{p_{i,t}}.
\end{aligned}$$

It follows that

$$\frac{\partial y_{i,t}^d(p_{i,t})}{\partial p_{i,t}} = -\varepsilon_{i,t}^d \frac{y_{i,t}^d}{p_{i,t}},$$

where $\varepsilon_{i,t}^d$ is the price elasticity of demand for good i . Define the final consumption fraction of total sales of good i as $v_{i,t}$:

$$v_{i,t} := \frac{p_{i,t} c_{i,t}}{p_{i,t} c_{i,t} + \sum_{j=1}^N p_{i,t} x_{ji,t}}.$$

We can then express the elasticity using this final consumption fraction:

$$\varepsilon_{i,t}^d = 1 + (\xi - 1) v_{i,t}.$$

Substituting this into the derivative of $y_{i,t}^d(p_{i,t})$, we can derive the optimal price as the following fixed-point equation (1.18):

$$p_{i,t} = \left(1 + \frac{1}{(\xi - 1)v_{i,t}}\right) (1 + i_t^L) \mu_{i,t|I_t}^{A-1} \frac{\Phi(q_{i,t} - \sigma_{i,t})}{\Phi(q_{i,t})} m c_{i,t}.$$

I.1.3 Firm's Optimal Working Capital

The optimal working capital problem is a constrained optimization problem:

$$\max_{b_{i,t}, n_{i,t}, \{x_{ij,t}\}_{j=1}^N} \mathbb{E} \left[\Pi_{i,t}^y \mid \mathcal{F}_t \right],$$

subject to

$$\text{Working capital constraint: } W_t n_{i,t} + \sum_{j=1}^N p_{j,t} x_{ij,t} \leq b_{i,t};$$

$$\text{Borrowing-capacity constraint: } b_{i,t} \leq \bar{b}_{i,t}.$$

Consider the objective function. Given the information set $\mathcal{F}_t$, there is no uncertainty. The objective function is therefore deterministic as the maximizer function. Note that if the objective function is constantly 0, it is indifferent for firms to choose any level of the choice variables. Therefore, we impose such that when the objective function is 0, the choice variables will be optimal such that they maximizes the operating surplus $p_{i,t} y_{i,t} - (1 + i_t^L) b_{i,t}$.

Denote the Lagrange multipliers for the working capital constraint and the borrowing-capacity constraint as $\lambda_{1i,t}^y$ and $\lambda_{2i,t}^y$, respectively. The FOCs are

$$[b_{i,t}] : -(1 + i_t^L) + \lambda_{1i,t}^y - \lambda_{2i,t}^y = 0; \tag{I.5}$$

$$[n_{i,t}] : p_{i,t} \alpha_{i,t} \frac{y_{i,t}}{n_{i,t}} - \lambda_{1i,t}^y W_t = 0; \tag{I.6}$$

$$[x_{ij,t}] : p_{i,t} \omega_{ij,t} \frac{y_{i,t}}{x_{ij,t}} - \lambda_{1i,t}^y p_{j,t} = 0, \forall j = 1, \dots, N. \tag{I.7}$$

A first observation is that the working capital constraint is binding.

Lemma I.3 (Binding Working-Capital Constraint). *The working-capital constraint (1.2) binds in any interior solution with finite $b_{i,t}$.*

Proof. From the FOC with respect to $b_{i,t}$,

$$\lambda_{1i,t}^y - \lambda_{2i,t}^y = (1 + i_{i,t}^L).$$

Since $1 + i_{i,t}^L > 0$, we have

$$\lambda_{1i,t}^y = \lambda_{2i,t}^y + (1 + i_{i,t}^L) > 0.$$

By complementary slackness for the working-capital constraint,

$$\lambda_{1i,t}^y \left[b_{i,t} - W_t n_{i,t} - \sum_{j=1}^N p_{j,t} x_{ij,t} \right] = 0.$$

Since $\lambda_{1i,t}^y > 0$, it follows that

$$b_{i,t} = W_t n_{i,t} + \sum_{j=1}^N p_{j,t} x_{ij,t}.$$

□

Second, take ratio of (I.6) and (I.7), we get

$$\frac{W_t n_{i,t}}{p_{j,t} x_{ij,t}} = \frac{\alpha_{i,t}}{\omega_{ij,t}}.$$

Combining with the binding working capital constraint, we get the cost-share conditions:

$$n_{i,t} = \frac{\alpha_{i,t} b_{i,t}}{W_t}, \quad x_{ij,t} = \frac{\omega_{ij,t} b_{i,t}}{p_{j,t}}, \forall j = 1, \dots, N.$$

I.1.4 Proof of Proposition 1.1

Proof. First, we introduce the following notations to rewrite the optimal pricing equation (1.18) in vector form. Let $\mathbf{p}_t = (\log p_{1,t}, \dots, \log p_{N,t})'$ and $\mathbf{q}_t = (q_{1,t}, \dots, q_{N,t})'$. Since the final consumption fraction $v_{i,t}$ is a function of prices, define

$$\mathbf{\Gamma}_t(\mathbf{p}_t) = \left(\log \left(1 + \frac{1}{(\xi - 1)v_{1,t}(\mathbf{p}_t)} \right), \dots, \log \left(1 + \frac{1}{(\xi - 1)v_{N,t}(\mathbf{p}_t)} \right) \right)'.$$

Define the log marginal cost vector being $\mathbf{mc}_t(\mathbf{p}_t)$. Recall that we have

$$mc_{i,t} = \frac{W_t^{\alpha_{i,t}} \prod_{j=1}^N p_{j,t}^{\omega_{ij,t}}}{\alpha_{i,t}^{\alpha_{i,t}} \prod_{j=1}^N \omega_{ij,t}^{\omega_{ij,t}}}.$$

Therefore, with the unnormalized network matrix Ω_t , we can write $\mathbf{mc}_t(\mathbf{p}_t)$ as

$$\mathbf{mc}_t(\mathbf{p}_t) = \Omega_t \mathbf{p}_t + \mathbf{w}_t,$$

where $\mathbf{w}_t$ is the vector of log coefficients and wage. We can also write $\mathbf{q}_t$ as

$$\mathbf{q}_t = \mathbf{q}_t(\mathbf{p}_t) := \Sigma_t^{-1} \left(\mathbf{a} + \underbrace{(I_N - \Omega_t)}_{=: \mathcal{L}_t} \mathbf{p}_t - \mathbf{i}^L_t - \mathbf{w}_t \right),$$

where $\Sigma_t = \text{diag}(\sigma_{1,t}, \dots, \sigma_{N,t})$ is the diagonal matrix of uncertainty, $\mathbf{a} = (\bar{a}_1, \dots, \bar{a}_N)'$ is the vector of log productivity means, and $\mathbf{i}^L_t = \log(1 + i_t^L) \mathbf{1}_N$ is the vector of loan rates.

Denote the default-risk adjustment vector as $\mathbf{F}_t(\mathbf{q}_t)$ with entries $\log \frac{\Phi(q_{i,t} - \sigma_{i,t})}{\Phi(q_{i,t})}$.

Using matrix notation, we can rewrite the optimal pricing equation (1.18) as

$$\mathbf{p}_t = \Gamma_t(\mathbf{p}_t) + \mathbf{F}_t(\mathbf{p}_t) + \Omega_t \mathbf{p}_t + \mathbf{w}_t + \mathbf{i}^L_t - \mathbf{a} + \frac{1}{2} \Sigma_t^2 \mathbf{1}_N.$$

Under Assumption 1.1, we can further rewrite the equation as

$$\mathbf{p}_t = \mathcal{L}_t^{-1} \left(\Gamma_t(\mathbf{p}_t) + \mathbf{F}_t(\mathbf{p}_t) + \mathbf{w}_t + \mathbf{i}^L_t - \mathbf{a} + \frac{1}{2} \Sigma_t^2 \mathbf{1}_N \right) \quad (\text{I.8})$$

Note that equation (I.8) is a fixed-point equation for $\mathbf{p}_t$. Next, we try to show that the fixed-point exists and is unique. We will use the Banach fixed-point theorem. Next, we show that the right-hand side of equation (I.8) is a contraction mapping.

□